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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Ms. Mohammed Razia Alangir Banu
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Roll Number	2520090121
Branch	CS-IT
Course Outcome	CO-3

Case Study Topic: Tourist City Connectivity using BFS, DFS and Prim's Minimum Spanning Tree

Work Done Summary:

In this case study, a graph-based tourism network was developed for the TourPlan – Smart Tourism & Travel Itinerary Optimization System. Tourist cities were represented as graph vertices, and travel routes between cities were represented as weighted edges. BFS and DFS algorithms were implemented to explore city connectivity and traversal paths, while Prim's Algorithm was used to construct a Minimum Spanning Tree (MST) for identifying the minimum-cost tourism network. The implementation demonstrates efficient graph traversal and network optimization techniques used in real-world transportation and travel planning systems.

Implementation Details:

1. Created a graph representation of tourist cities using an adjacency matrix.
2. Added weighted edges to represent travel distances between cities. .
3. Implemented Breadth First Search (BFS) to explore connected tourist destinations level by level.
4. Implemented Depth First Search (DFS) to traverse all reachable cities recursively.
5. Applied Prim's Algorithm to construct a Minimum Spanning Tree (MST).
6. Calculated the minimum travel network cost connecting all tourist cities.
7. Displayed BFS traversal, DFS traversal, and MST results.
8. Verified graph connectivity and optimized route planning for tourism applications.

RESULT : SCREENSHOTS/OUTPUT

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=== TOURPLAN CITY CONNECTIVITY SYSTEM ===

BFS Traversal: 0 1 2 3 4

DFS Traversal: 0 1 2 4 3

Minimum Spanning Tree:
Hyderabad -> Warangal = 120 km
Warangal -> Vizag = 100 km
Warangal -> Araku = 200 km
Vizag -> Vijayawada = 150 km

Total MST Cost = 570 km
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This case study implements BFS, DFS, and Prim's Algorithm to analyze tourist city connectivity in the TourPlan system. Graph traversal techniques efficiently explore connected destinations, while Prim's Algorithm identifies the minimum-cost network connecting all cities. The results demonstrate the importance of graph algorithms in travel planning, transportation networks, and route optimization systems.

GitHub link:https://github.com/Jhanasri-thota/DSA_CASE-STUDIES

Student Signature	Faculty Signature